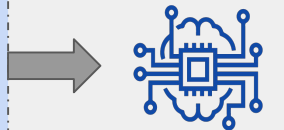


The Direct Perception Flow

[illegible]

Query: "What is the total gross amount"

System Prompt:
No Reasoning



Model
(Qwen3-VL)

Direct Perception Path

Output Group (Sampling)

 O_1

```
"Answer": "684.93",
"box_2d": [x1, y1, x2, y2]
```

 O_2

```
"Answer": "684.93",
"box_2d": [x1, y1, x2, y2]
```

O.

"Answer": "684.03",
box_2d": [x1, y1, x2, y2]

Advantage Estimation

$$\frac{r - \mu}{\sigma}$$

GRPO Update (π^*)

Aggregated Rewards

The Gated Reward Mechanism

Reasoning-Free Rewards

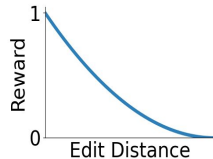
Format Constraint (Binary Switch)

01

✓ or ✗

Is Valid JSON?

Semantic Precision (Quadratic Curve) Text Matching (ANLS)



The Localization Gate (The “Cliff-Edge”) Gated Spatial Reward (Threshold = 0.6)

